UCCN2243 Internetworking Principles and Practices

UCCN2003 TCP/IP Internetworking

This group assignment consists of 2 questions. The due date of the assignment is on week 10, Wednesday, 8 April 2026 before 5pm.

This is a group assignment, **group of 3 students MINIMUM OR group of 4 students MAXIMUM.**

Group members MUST be from the same Practical LAB session. Marks will be DEDUCTED if the assignment group is LESS than 3 students (UNLESS pre-assigned by the instructor)

The report cover page should include:
1) Group member's names,
2) Student ID
3) Practical group (eg. P1, P2, P3)
4) Programme (eg. CN, CT, CS, IA, IR)
5) Subject code.

This work should be submitted in softcopy. (Assignment submission link will be provided in WBLE)

WARNING: You are STRONGLY advised NOT to copy or allow your report to be copied by your friends. Heavy penalty will be enforced on those found to have committed this offence.

Sufficient referencing (IEEE style) are required for contents cited from other resources.

Assignment Q1 (10 marks)

Refer to the network in Q1.pkt

- Company AA network is assigned the public address 201.1.1.1. It has an offsite network which hosts the Remote DHCP Server0 that could be access with the public address 201.1.1.2.

- Configure the remaining gateways with the last usable address.

- Configure Port Address Translation (PAT) on Router0 and Router1 to enable the end devices to access the public network.

- Configure default route on both routers.

- Configure Router0 and Router1 to enable PC0, PC1 and PC2 to access the DHCP service from Server0.

a) Show the running configurations of Router0 and Router1. Highlight the relevant parts and provide brief explanation.

b) Show that PC0, PC1 and PC2 have successfully obtained their IP addresses from the Remote DHCP Server0. Include descriptions.

c) Could PC0, PC1 and PC2 DIRECTLY ping the Remote DHCP Server0? Explain WHY / WHY NOT.

You may use “print screen” to present the results.

Assignment Q2 (15 marks)

Use GNS3 to build the network as shown in the Figure Q2.

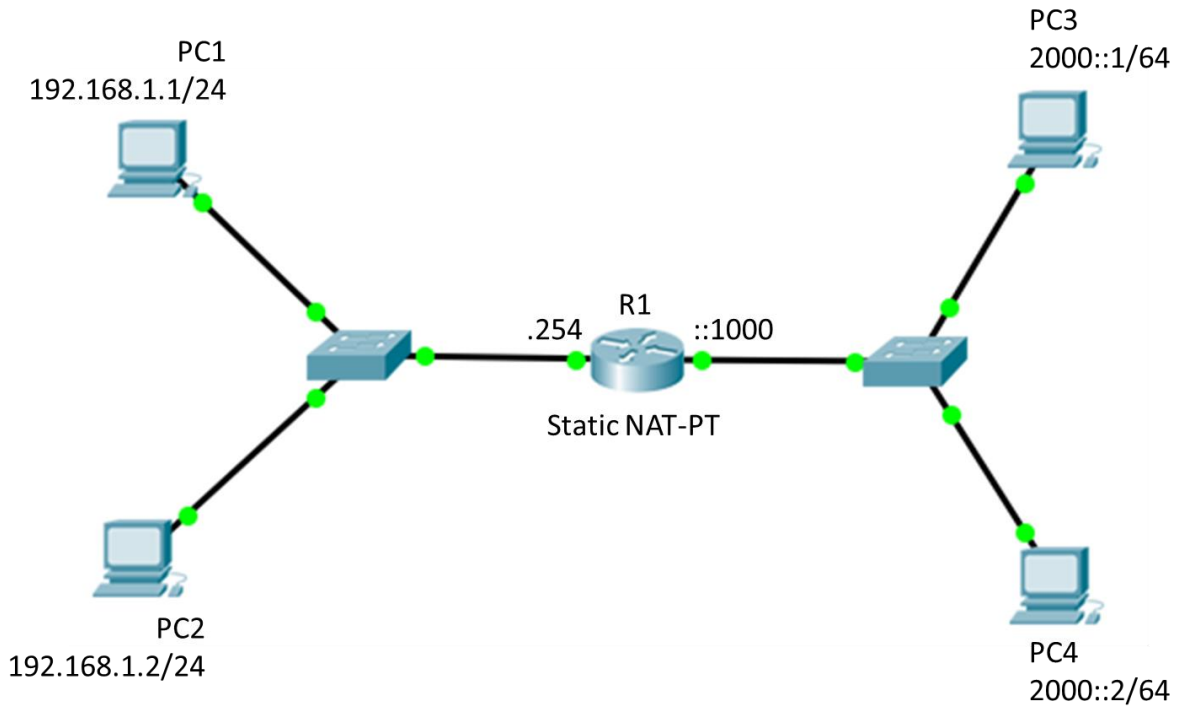

Figure Q2

Assume all the subnet masks to be /64 (for IPv6) and /24 (for IPv4) respectively.

Assign the gateway address for PC1 and PC2 to be 192.168.1.254

Assign the gateway address for PC3 and PC4 to be 2000::1000

a) Configure Static NAT-PT in router R1 to enable both subnets to be able to communicate with each other.

- Let the IPv4 addresses of PC1 and PC2 be translated to 2001::1 and 2001::2 respectively
- Let the IPv6 addresses of PC3 and PC4 be translated to 192.168.2.1 and 192.168.2.2 respectively

b) Show (include descriptions) the relevant running configuration for R1.

c) Show (include descriptions) how PC1/PC2 would ping PC3/PC4.

d) Use Wireshark (integrate it to GNS3) and show (include descriptions) that IP addresses of the respective packets are translated from IPv4 to IPv6 and vice-versa between PC1/PC2 and PC3/PC4.

You may use “print screen” to present the results.

Hints:

GNS3 is an opensource program which you can download for free.

The router’s image file as well as the image file for the end host (you may use the Qemu images) are made available to you. **NOTE: You may use the image files needed for GNS3 from the links provided in WBLE.**

You will need to find out how to integrate Wireshark into GNS3 in order for it to capture and observes the ping packets.